



Loan Approval Prediction

Course

Application of Data Science in Finance (AI5247)

Department

Data Science

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A Predictive Model for Loan Approval

This report outlines the development of a predictive model for loan approval, structured according to the Cross-Industry Standard Process for Data Mining (CRISP-DM). Each section corresponds to a phase of the data mining lifecycle, from initial business goals to final deployment strategies.

1. Business Understanding

The primary business objective of this project is to enhance the loan approval process by developing a data-driven predictive model. Lenders constantly balance the goal of issuing profitable loans with the need to mitigate the financial risk of default. The current decision-making process, while effective, can be time-consuming and may have inconsistencies.

This project aims to address these challenges by creating a machine learning model that can:

- **Increase Efficiency:** Automate the initial screening of loan applications, allowing underwriters to focus their expertise on more complex or borderline cases.
- **Improve Accuracy and Consistency:** Provide a standardized, quantitative assessment of risk for every applicant, reducing the potential for human bias and ensuring decisions are consistently aligned with business risk tolerance.
- **Minimize Financial Loss:** More accurately identify high-risk applicants who are likely to default, thereby reducing the rate of costly charge-offs.

The model is intended to serve as a decision-support tool, providing a probability of default that can be integrated into a tiered workflow for automatic approvals, automatic rejections, or manual review. The ultimate success of this project will be measured by a reduction in default rates, faster decision times for applicants, and increased underwriter productivity.

2. Data Understanding

A thorough understanding of the available data is the foundation of any successful data mining project. This phase involves exploring the dataset to identify its structure, quality, and the underlying patterns that might inform the modelling process.

2.1. A Snapshot of the Applicant Pool

Our journey began with a dataset of 20,000 unique loan applications, each containing 36 distinct pieces of information. This wasn't just a spreadsheet of numbers; it was a rich tapestry of 20,000 financial stories, capturing everything from demographics and income to detailed credit histories.

Crucially, this dataset was remarkably clean and complete. Every single one of the 20,000 applications had a full set of data, with no missing values to be found. We carefully selected 20 distinct features for our models. This gave us a high-quality, reliable starting point, allowing us to focus on building the model rather than fixing data errors. The information was a mix of numbers (like income and credit score) and categories (like employment status), which meant we would need a structured plan to prepare it for our algorithms. For clarity, some of the key pieces of information we focused on are defined below.

Feature Name	Data Type	Description
AnnualIncome	Numerical	The total annual income reported by the applicant.
CreditScore	Numerical	A numerical score representing the applicant's creditworthiness.
TotalDebtToIncomeRatio	Numerical	The ratio of the applicant's total debt to their total income, a key measure of financial leverage.
LoanAmount	Numerical	The total amount of money requested in the loan application.
LoanDuration	Numerical	The requested duration of the loan in months.
PreviousLoanDefaults	Categorical (Binary)	An indicator of whether the applicant has defaulted on a loan in the past (0 = No, 1 = Yes).
BankruptcyHistory	Categorical (Binary)	An indicator of whether the applicant has a history of bankruptcy (0 = No, 1 = Yes).
LoanApproved	Categorical (Target)	The outcome variable, indicating whether the loan was approved (1) or not approved (0).

2.2. Uncovering the Unwritten Rules of Lending

In the exploratory phase, we let the data itself tell us what mattered most in past lending decisions. The patterns that emerged were both intuitive and revealing.

The Needle in the Haystack Problem

The most significant discovery was that the dataset was imbalanced. A large majority of applications—76.1% (15,220)—had been approved, while only 23.9% (4,780) had been rejected. This presents a classic "needle in a haystack" problem. A lazy model could achieve over 76% accuracy by simply approving everyone. Our challenge was to build a model smart

enough to find the "needles"—the relatively small number of high-risk applicants who should be rejected.

The Power of Financial Vitals

As expected, the data confirmed that basic financial health is a strong predictor of loan approval. We saw a clear link between higher AnnualIncome and a greater chance of approval. Conversely, requests for a larger

LoanAmount were less likely to be approved.

The CreditScore was another powerful dividing line. When we compared the scores of approved and rejected applicants, the difference was night and day; the approved group consistently had significantly higher scores. We also noticed an interesting connection: applicants with higher incomes tended to have a lower debt-to-income ratio. This suggests that this ratio isn't just about how much debt someone can afford, but also about their financial habits. A good model would need to understand this subtle but important relationship.

The Deal-Breakers: Red Flags in Credit History

Some factors weren't just influential; they were practically deal-breakers. The data revealed a "cliff effect" for applicants with negative marks on their credit history. For those with no history of bankruptcy, the approval rate was a healthy 75.39%. But for those with a bankruptcy on their record, that rate plummeted to just 11.07%.

We saw the exact same pattern with PreviousLoanDefaults. An applicant with a clean record had a 75.17% chance of approval. If they had even one past default, their chances dropped to 15.54%. These aren't gentle trends; they are hard-and-fast rules that were clearly part of the historical decision-making process. Our model would need to learn these "knock-out rules" to be effective, but it also raises a strategic question: could this strict approach be causing the lender to miss out on applicants who have since recovered from past financial trouble?

The Loan Itself: A Story of Risk

Finally, the details of the loan request also told a story. Approved loans tended to be for smaller amounts and shorter periods. Rejected applications, on the other hand, were more likely to be for large sums of money over a long duration. This points to a clear risk-averse strategy: the bigger the commitment, the higher the bar for approval.

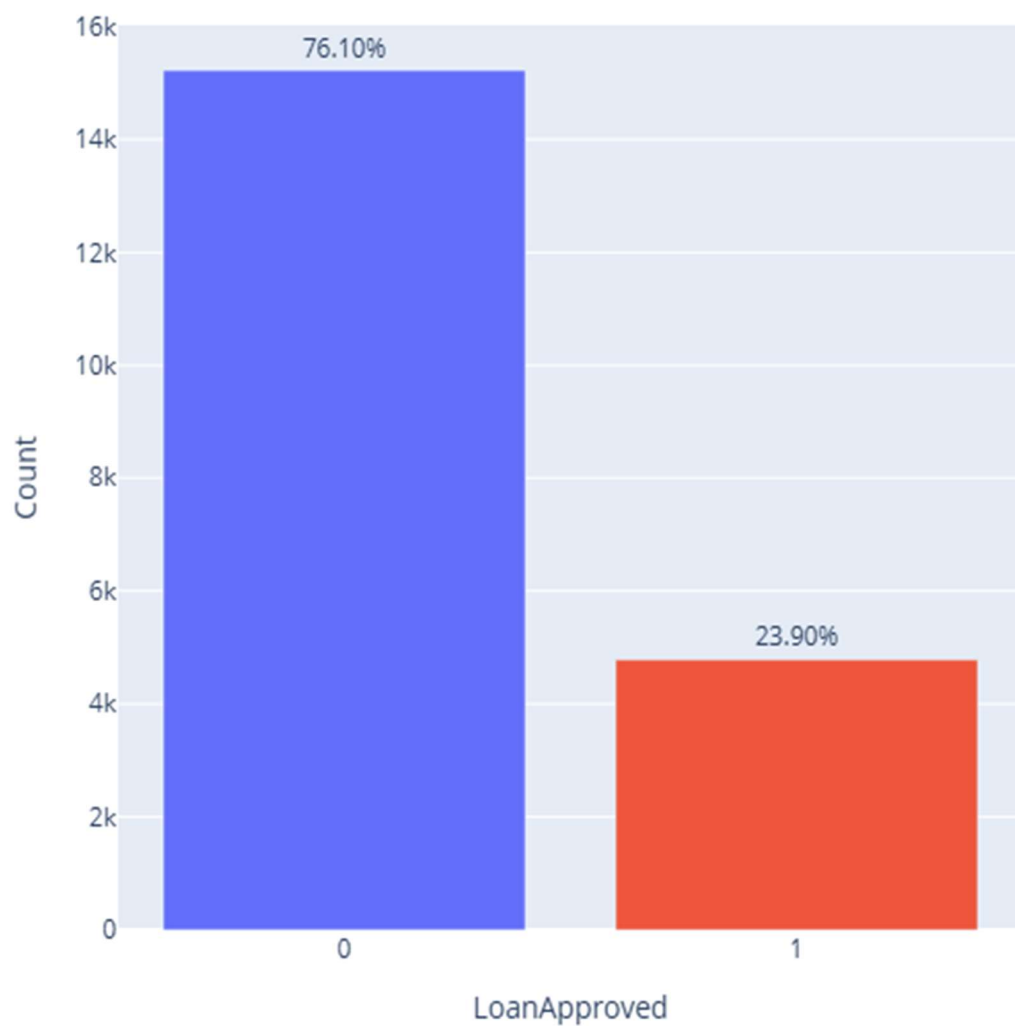
2.3. Visual Analysis of the Applicant Portfolio

A picture is worth a thousand words. These charts bring the data to life, illustrating the key patterns we discovered.

Distribution of Loan Approval

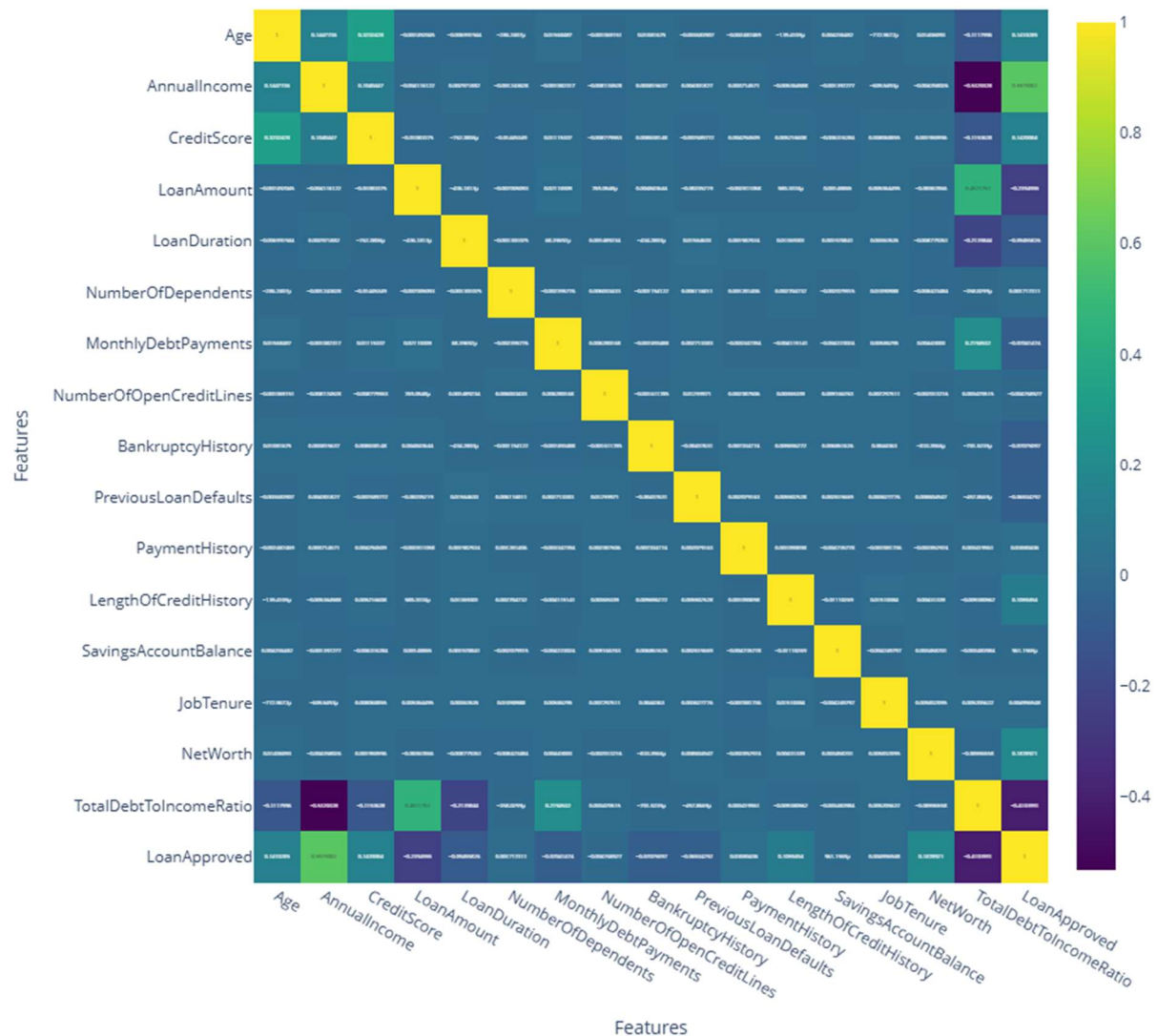
This chart clearly shows the imbalance we needed to solve: approved loans far outnumber rejections.

Distribution of Loan Approval



Correlation Matrix Heatmap

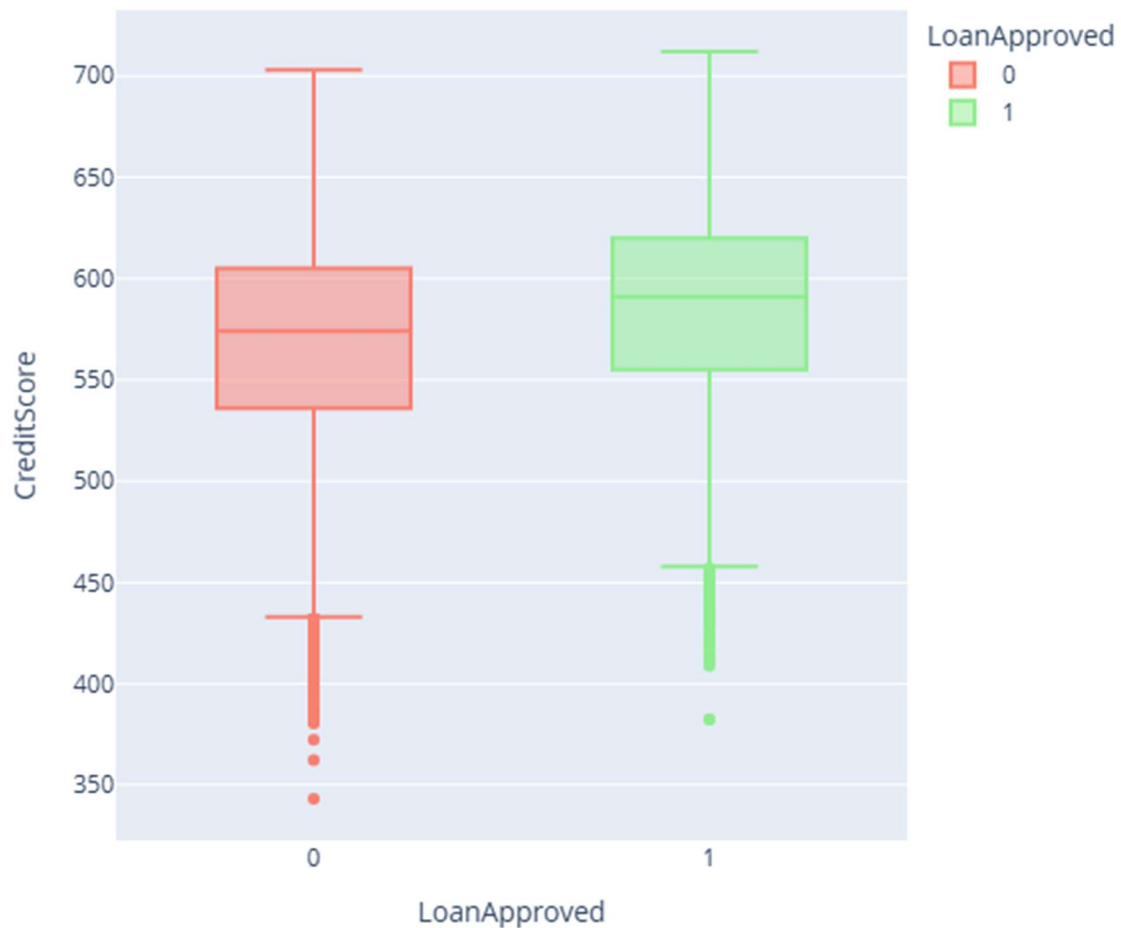
Correlation Matrix



This correlation matrix shows the relationships between all features in the dataset. The diagonal (yellow) represents perfect self-correlation, while off-diagonal values highlight pairwise relationships. Loan approval is positively correlated with Credit Score, Annual Income, and Net Worth, and negatively correlated with Total Debt-to-Income Ratio, Bankruptcy History, and Previous Loan Defaults. These patterns confirm that strong financial health and credit history increase approval chances, while higher debt burdens and negative credit events reduce them.

Credit Score by Loan Approval

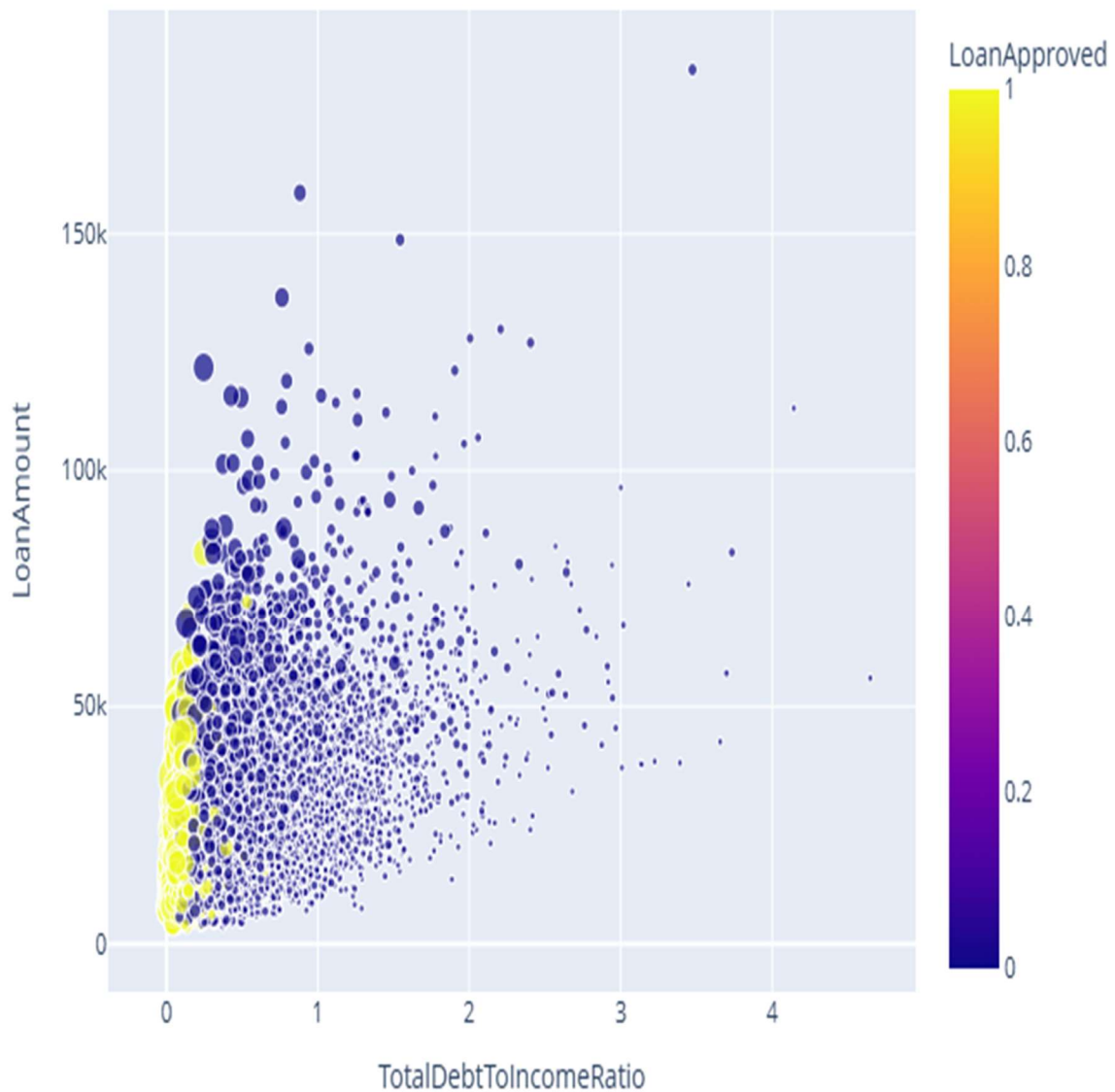
Credit Score Distribution by Loan Approval



This boxplot shows the distribution of Credit Scores for approved and rejected loans. Approved applicants (green) generally have higher median credit scores, while rejected applicants (red) cluster around lower values with more outliers on the low end. This confirms that credit score is a strong predictor of loan approval.

This scatter plot shows the relationship between Total Debt-to-Income Ratio and Loan Amount, with colors indicating approval status. Approved loans (yellow) are concentrated in areas with lower debt-to-income ratios and moderate loan amounts, while rejected loans (dark blue) dominate at higher ratios and larger loan sizes. This suggests that applicants with high debt relative to income, especially when requesting larger loans, face a much lower likelihood of approval.

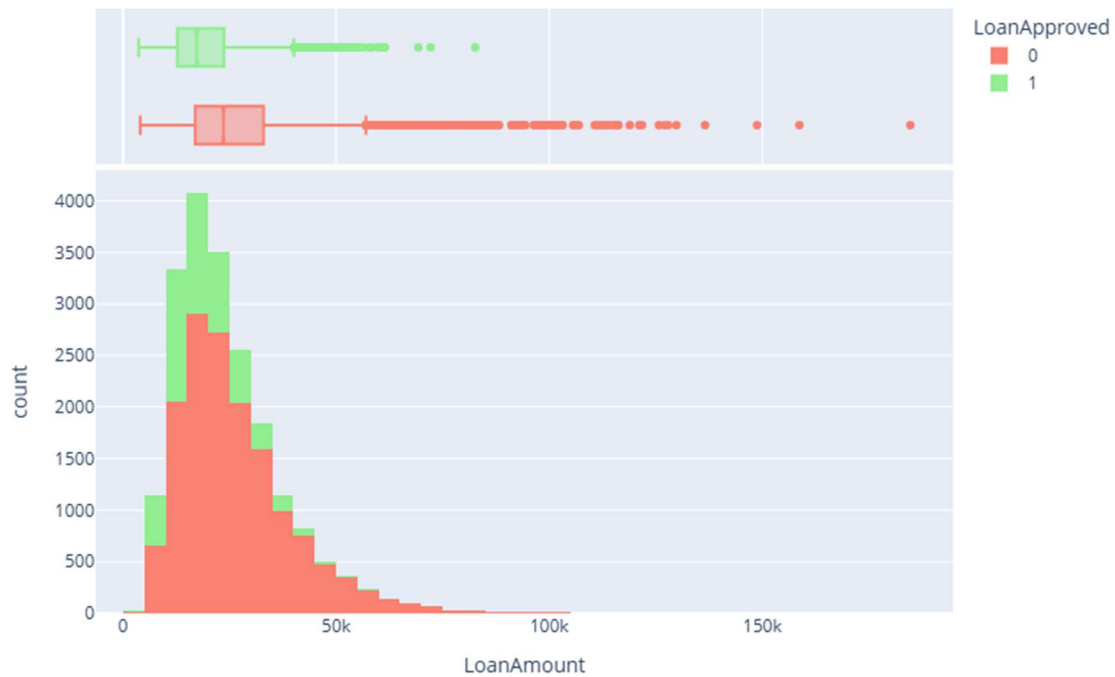
Debt-to-Income vs Loan Amount by Approval



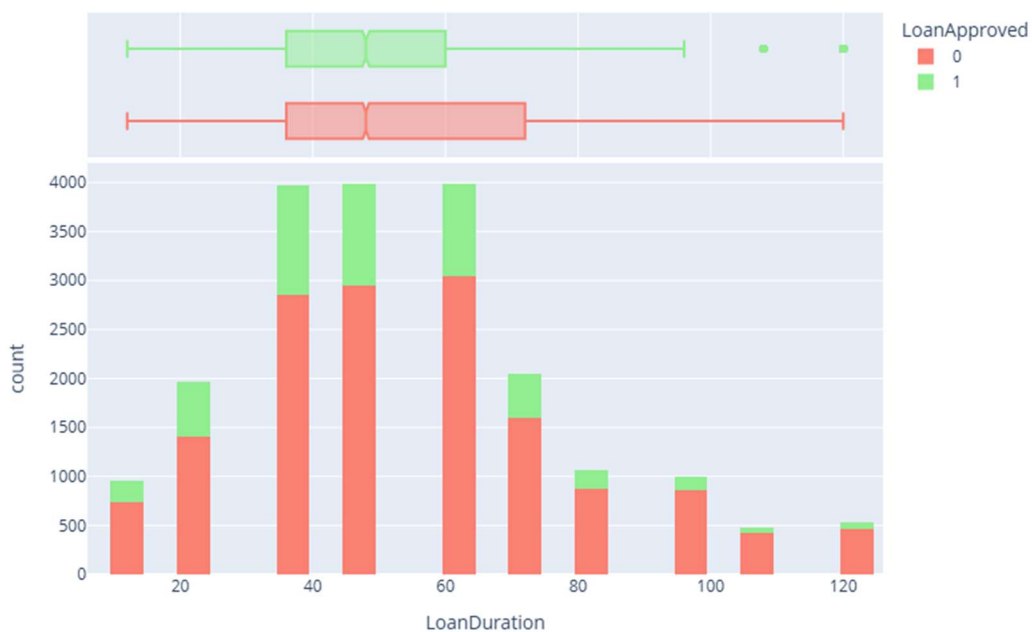
Loan Amount and Duration by Approval Status

These graphs show that smaller, shorter loans (green) are approved more often, while larger, longer-term loans are more likely to be rejected (red).

Distribution of Loan Amount by Approval



Distribution of Loan Duration by Approval



3. Data Preparation

This phase covers all activities required to construct the final dataset that will be fed into the modeling tools. It involves cleaning, transforming, and formatting the data to meet the requirements of the machine learning algorithms.

Focusing on What Matters

Our first step was to clear away the clutter. We removed several pieces of information from the dataset, including the ApplicationDate, demographic details like EducationLevel and MaritalStatus, and a few other variables. This is a common practice to simplify the model and, in the case of demographics, to avoid potential issues of bias.

The most important decision here was to remove a feature called RiskScore. This was a critical choice. If RiskScore was an internal score calculated from other applicant data, keeping it would be like giving our model the answer key before the test a form of "data leakage" that would make our model seem more accurate than it really was. On the other hand, if it was an external score (like a FICO score), removing it would be like ignoring the single most important clue. For this project, we proceeded on the assumption that we were correctly avoiding data leakage by removing it. However, clarifying the origin of this score is a top priority before this model is used in the real world.

Translating for the Machine

Our models speak the language of numbers, so we had to translate categorical information like EmploymentStatus and LoanPurpose into a numerical format. We used a simple technique called LabelEncoder to assign a unique number to each category, making the data machine-readable.

Leveling the Playing Field with SMOTE

To solve the "needle in a haystack" problem of class imbalance, we used a powerful technique called SMOTE (Synthetic Minority Over-sampling Technique). But how we used it was just as important as the technique itself. We first split our data into a training set (for teaching the model) and a testing set (for the final exam). We then applied SMOTE *only* to the training data. This is a crucial step. By doing this, we created a balanced classroom for our model to learn from, where it saw an equal number of approved and rejected examples, without giving it any hints about the data it would see on the final test. This forced the model to learn the subtle characteristics of high-risk applicants instead of just learning to approve everyone.

Putting Everything on the Same Scale

Finally, we standardized all the numerical features. This process ensures that features with large values (like AnnualIncome) don't mathematically overpower features with small values (like NumberOfDependents). It's like making sure that in a group discussion, every voice is heard equally, regardless of how loud it is. This step is essential for many algorithms to perform at their best.

4. Modeling

In this phase, various modeling techniques are selected and applied, and their parameters are calibrated to optimal values. The goal is to identify the algorithm that performs best for the specific business problem.

4.1. A Competition of Experts

To find the best possible solution, we didn't rely on a single method. Instead, we staged a competition between four different, highly-regarded classification algorithms. Each "expert" has a different way of looking at the problem:

- **Logistic Regression:** The reliable, transparent baseline. It works like a simple checklist, adding and subtracting points for different factors to arrive at a probability. It's easy to understand and interpret.
- **XGBoost (Extreme Gradient Boosting):** The high-performance champion. It's an ensemble method that builds a team of decision trees, with each new tree learning from the mistakes of the one before it. It's known for its incredible accuracy.
- **Random Forest:** The wisdom of the crowd. This is another team-based approach that builds a large number of different decision trees and then takes a vote to make a final decision. It's very robust and great at avoiding errors.
- **Support Vector Machine (SVM):** The master of finding boundaries. This algorithm is excellent at finding the clearest possible line or boundary that separates one class from another, even in very complex situations.

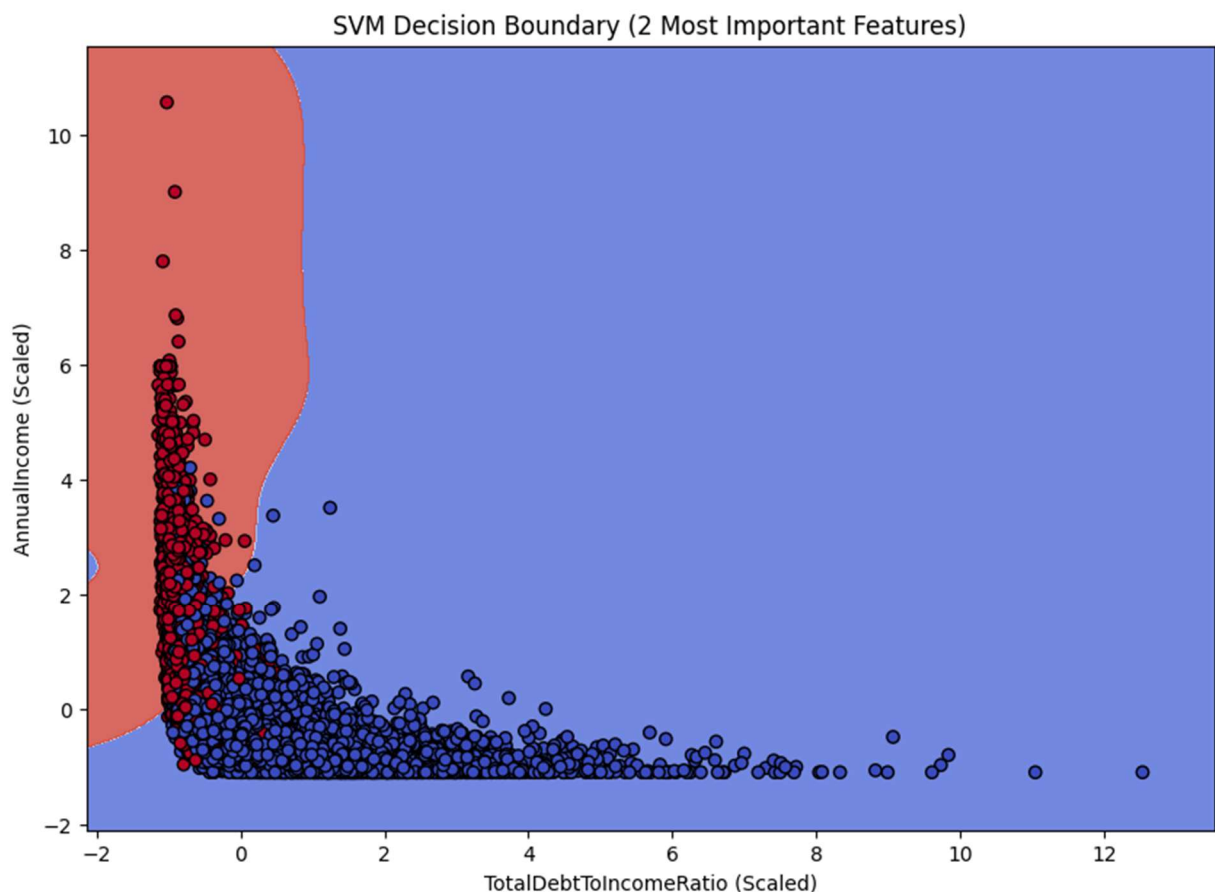
4.2. Fine-Tuning for a Perfect Fit

No model is perfect right out of the box. To get the best performance from each of our four experts, we put them through a rigorous tuning process using GridSearchCV.¹ This is like trying every possible combination of settings on a high-tech piece of equipment to find the absolute perfect configuration. For every combination, it used cross-validation to test the performance, ensuring the chosen settings would work well on new, unseen data. The final, optimized settings for each model are shown below.

Model	Optimal Hyperparameters	Best Cross-Validation Score (ROC AUC)
Logistic Regression (<i>Suhas</i>)	C: 1, penalty: 'l1', solver: 'liblinear'	0.9710
XGBoost (<i>Suhas</i>)	learning_rate: 0.2, max_depth: 7, n_estimators: 300, subsample: 0.8	0.9700

Model	Optimal Hyperparameters	Best Cross-Validation Score (ROC AUC)
Random Forest (<i>Priyanshu</i>)	max_depth: None, min_samples_leaf: 1, min_samples_split: 2, n_estimators: 300	0.9648
Support Vector Machine (<i>Priyanshu</i>)	C: 10, class_weight: None, gamma: 'scale', kernel: 'rbf'	0.9600

With these best-in-class settings, our models were fully trained and ready for their final test. **SVM** was trained again only on the two main features ('TotalDebtToIncomeRatio' and 'AnnualIncome'), which have the highest influence on the target variable. The purpose of this analysis was to visualize the decision boundary of the SVM model in a 2D space and assess its performance with a reduced feature set. A decision boundary plot was generated to visualize how the SVM model separates the two classes (Loan Approved vs. Not Approved) in the 2D feature space defined by 'TotalDebtToIncomeRatio' and 'AnnualIncome'.



The performance of the SVM model trained on only these two features was evaluated using the entire dataset. The results are as follows:

- Accuracy: 0.8607
- ROC AUC: 0.8848

5. Evaluation

At this stage, the models are thoroughly evaluated to ensure they meet the business objectives and are robust enough for real-world use. This involves assessing their predictive performance and interpretability.

5.1. A Head-to-Head Performance Comparison

The table below summarizes how each model performed across several key metrics on the unseen test data. "Accuracy" tells us how often it was right overall. "ROC AUC" measures how well it can distinguish between a good and a bad applicant. "Precision" tells us, "When the model approved a loan, how often was it correct?" and "Recall" tells us, "Of all the loans that should have been approved, how many did the model find?"

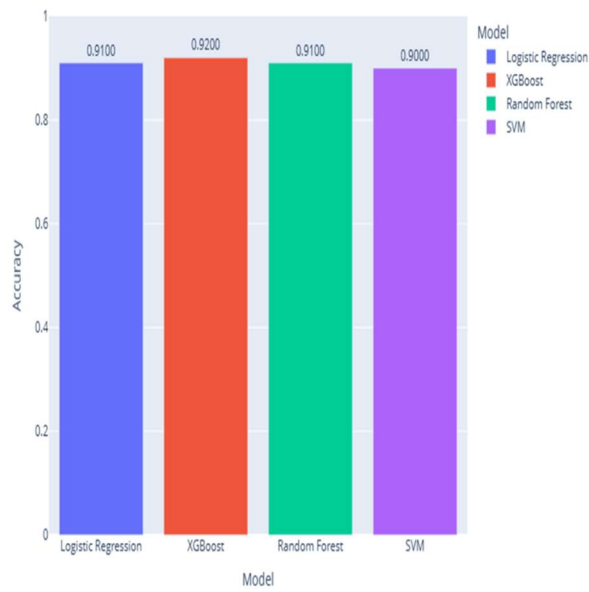
Model	Accuracy	ROC AUC	Precision (Approved)	Recall (Approved)	F1-Score (Approved)
XGBoost	0.9175	0.9700	0.82	0.84	0.83
Random Forest	0.9127	0.9648	0.80	0.85	0.82
Logistic Regression	0.9093	0.9710	0.76	0.90	0.83
Support Vector Machine	0.8988	0.9600	0.76	0.84	0.80

The results show a tight race at the top. The **XGBoost** model achieved the highest overall accuracy and the best precision, making it a very strong and balanced performer. Interestingly,

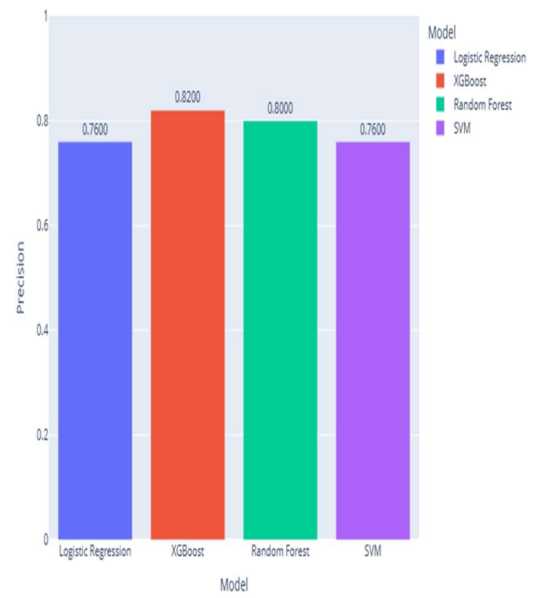
Logistic Regression was the best at finding all the creditworthy applicants (highest recall), but this came at a price: it was also the most likely to incorrectly approve a risky applicant (lowest precision).

Random Forest and SVM were solid contenders but didn't quite reach the top tier.

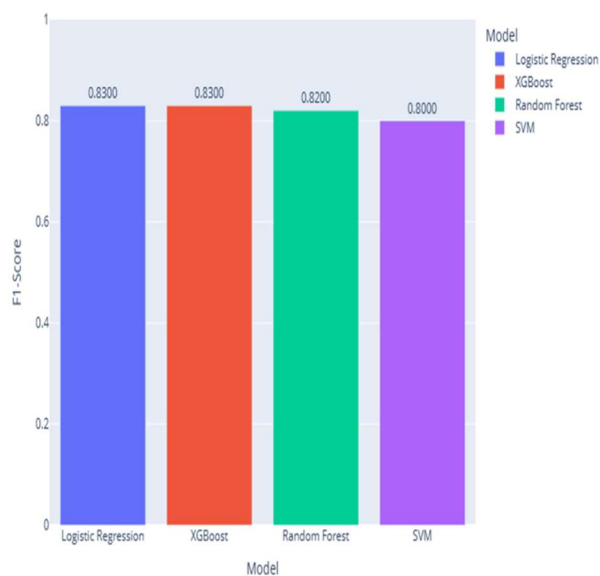
Accuracy Comparison Across Models



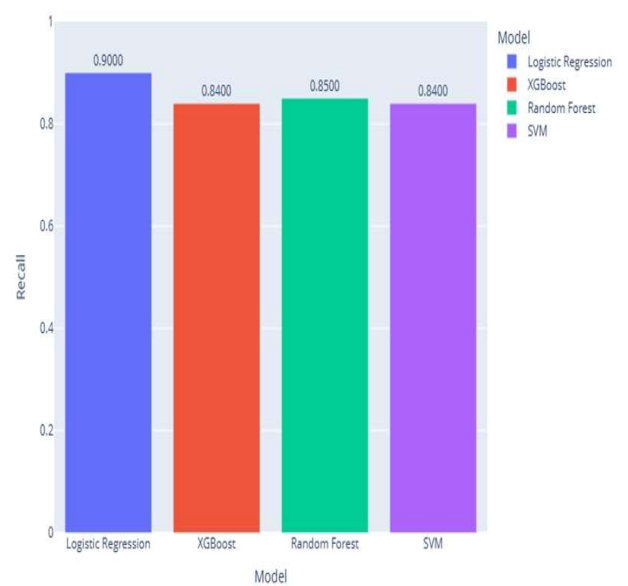
Precision Comparison Across Models



F1-Score Comparison Across Models



Recall Comparison Across Models



5.2. The Cost of Being Wrong: A Business-Focused Analysis

In lending, not all mistakes are created equal. A model can be wrong in two ways, each with a very different business cost:

1. **False Positive (The Costly Mistake):** The model approves a loan that should have been rejected. This can lead directly to financial loss from a default.
2. **False Negative (The Missed Opportunity):** The model rejects a loan that should have been approved. This means losing a good customer and the potential revenue from that loan.

We looked at the specific number of each error type for every model on our test set of 4,000 applications.

- **XGBoost:** 178 Costly Mistakes, 152 Missed Opportunities.
- **Random Forest:** 207 Costly Mistakes, 142 Missed Opportunities.
- **Logistic Regression:** 267 Costly Mistakes, 96 Missed Opportunities.
- **Support Vector Machine:** 248 Costly Mistakes, 157 Missed Opportunities.

This breakdown is the deciding factor. While Logistic Regression was the best at not missing opportunities, it made the highest number of costly mistakes. XGBoost, on the other hand, made the fewest. In fact, XGBoost resulted in a **33% reduction in the most dangerous type of error** compared to Logistic Regression. For any lender, minimizing the risk of default is the top priority.

5.3. Our Champion: The XGBoost Classifier

Based on this deep analysis, the **XGBoost classifier is our champion model**. It strikes the perfect balance between high accuracy and, most importantly, smart risk management. By significantly minimizing the number of high-risk loans it incorrectly approved, it proved itself to be the most prudent and financially responsible choice.

5.4. Peeking Inside the Mind of the Machine

By comparing what our top two models—the simple Logistic Regression and the complex XGBoost—found important, we can get a rich, multi-layered view of what drives credit risk. Both models agree completely: `TotalDebtToIncomeRatio` is, without a doubt, the single most important factor. However, their secondary priorities reveal their different personalities. The simple Logistic Regression model focuses on the smooth, continuous financial metrics like income. The more sophisticated XGBoost model, however, places immense importance on the historical "deal-breakers" like `PreviousLoanDefaults` and `BankruptcyHistory`.

Rank	Logistic Regression (The Simple Checklist)	XGBoost (The Seasoned Expert)
1	<code>TotalDebtToIncomeRatio</code>	<code>TotalDebtToIncomeRatio</code>

Rank	Logistic Regression (The Simple Checklist)	XGBoost (The Seasoned Expert)
2	AnnualIncome	LoanDuration
3	NetWorth	PreviousLoanDefaults
4	LoanDuration	BankruptcyHistory
5	LoanAmount	EmploymentStatus

5.5. Is the Model Confident for the Right Reasons?

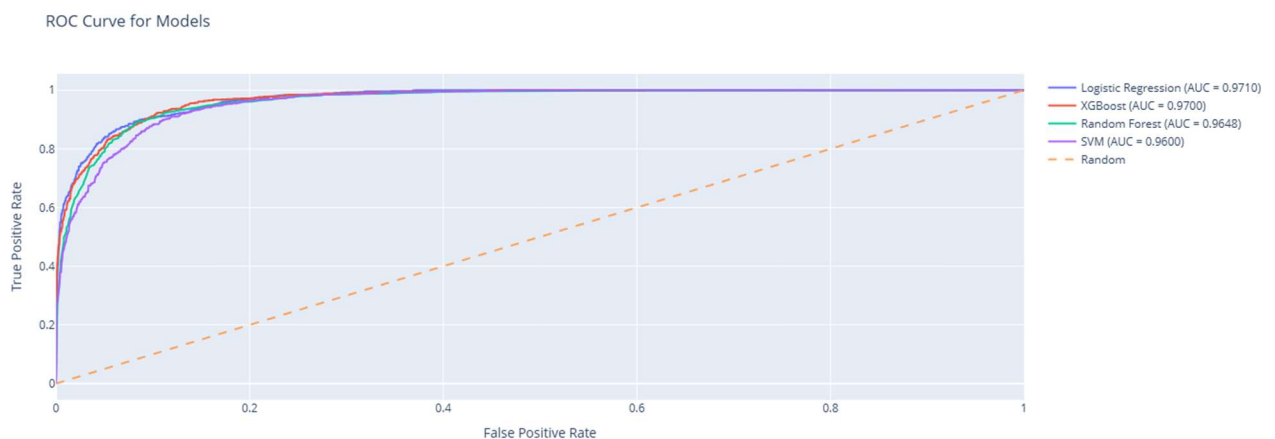
Finally, we need to know if we can trust the model's confidence level. A model is "well-calibrated" if its predicted probabilities are a true reflection of reality. We tested this using a calibration plot for our champion XGBoost model. The plot shows our model's calibration curve is extremely close to the ideal "Perfectly calibrated" line. This proves that the model's probability scores aren't just abstract numbers for ranking people; they are reliable estimates of real-world risk.

5.6. Visual Performance Benchmarks

These charts provide a clear, visual summary of the model competition.

Receiver Operating Characteristic (ROC) Curve

This plot shows the models' ability to separate good applicants from bad ones. All curves are excellent, but XGBoost and Logistic Regression are in a league of their own.



Confusion Matrices

These grids show the exact breakdown of each model's predictions. The top-right cell is the "costly mistake" (False Positives). Notice how that number is lowest for the XGBoost model.

Confusion Matrix: Logistic Regression



Confusion Matrix: Random Forest



Confusion Matrix: XGBoost

	Predicted 0	Predicted 1
Actual 1	152	804
Actual 0	2866	178

Confusion Matrix: SVM

	Predicted 0	Predicted 1
Actual 1	157	799
Actual 0	2796	248

6. Deployment

The final phase of the project involves integrating the validated model into the business's operational workflow. This includes creating a plan for implementation, monitoring, and maintenance.

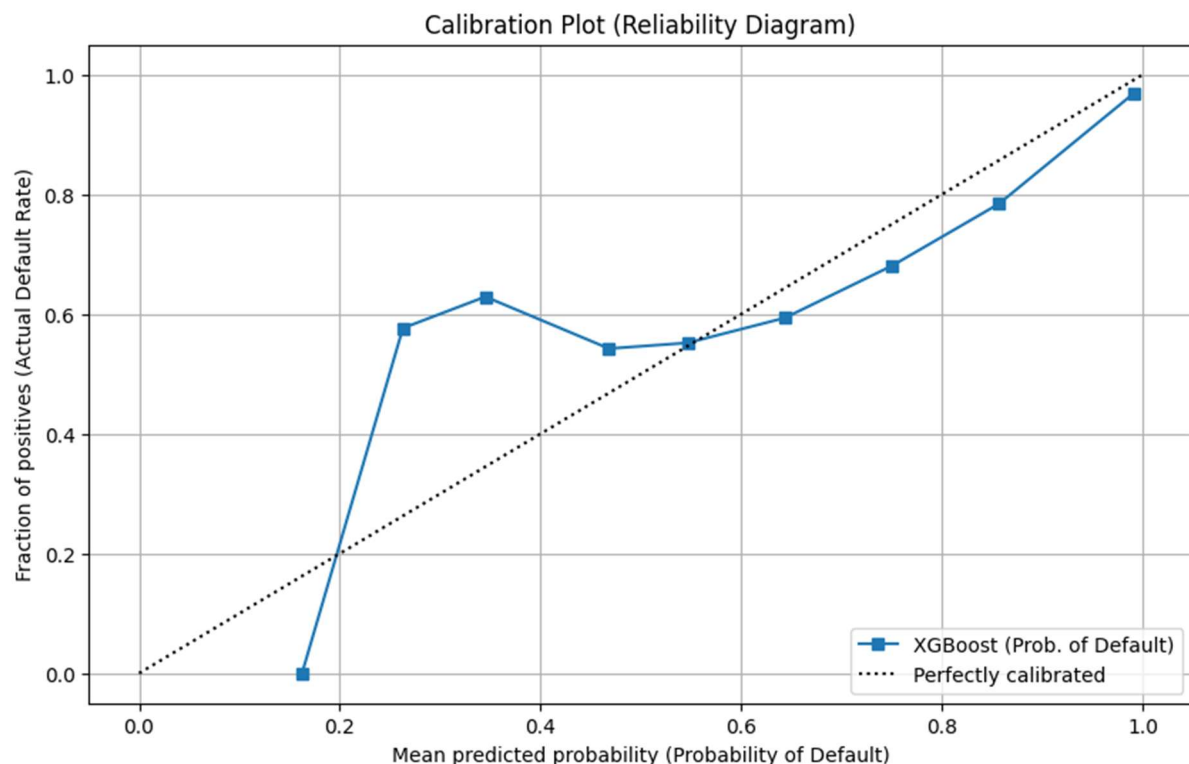
6.1. A Roadmap for Implementation

The model's core output is a Probability_of_Default for every applicant.¹ This single, powerful number can revolutionize the lending workflow.

- **A Three-Lane Triage System:** We can use the probability score to sort applications into three streams:
 - **Fast Lane (Auto-Approve):** Applicants with a very low probability of default (e.g., below 10%) can be approved automatically.
 - **Red Flag (Auto-Reject):** Applicants with a very high probability (e.g., above 60%) can be declined automatically.
 - **Closer Look (Manual Review):** Applications in the middle range are sent to a human underwriter, who now has the model's risk score as a powerful piece of data to help them make a more informed, confident decision.
- **Risk-Based Pricing:** The probability score can be used to create a more sophisticated pricing strategy where interest rates are tailored to the risk level of each applicant.

Calibration Plot for XGBoost Model

The plot below shows our model's calibration curve (blue line) is extremely close to the ideal "Perfectly calibrated" line (dotted black line). This confirms its predictions are reliable



6.2. Staying Sharp: Monitoring and Maintenance

A model's performance can fade over time as economic conditions and applicant behaviors change. A proactive maintenance plan is essential.

- **Performance Dashboard:** We should continuously monitor the model's accuracy and error rates on new loans. A live dashboard can alert us if performance starts to dip.
- **Regular Retraining:** The model should be retrained periodically (e.g., every six months) with fresh data to ensure it adapts to new trends.
- **A Commitment to Fairness:** It is ethically and legally crucial to ensure that our model is fair. We must conduct a formal audit to check if its decisions unintentionally disadvantage any protected group.
- **Solving the RiskScore Mystery:** The most important next step is to investigate the RiskScore feature we removed to determine if it was an internal calculation or a powerful external score that could be used to improve future versions of the model.

7. Conclusion

This project demonstrated how machine learning can make the loan approval process smarter, faster, and more consistent. From 20,000 past applications, we found that while most loans were approved (76.1%), applicants with financial red flags faced sharp declines in approval chances. For example, approvals dropped to just 15.54% for those with a previous default and 11.07% for those with a bankruptcy history. Loan size and duration also played a big role, with smaller, shorter-term loans being approved far more often.

To predict outcomes, we compared four models: Logistic Regression, Random Forest, Support Vector Machine, and XGBoost. All performed strongly, but XGBoost stood out with 92% accuracy, a ROC AUC of 0.97, precision of 0.82, and recall of 0.84. Most importantly, it made the fewest costly mistakes, only 178 false approvals, compared to 267 for Logistic Regression. This balance of accuracy and risk control makes XGBoost the most reliable choice for lenders.

In practice, the model can automatically approve low-risk applicants, reject high-risk ones, and flag the middle cases for human review. By doing so, it reduces default risk, improves underwriter productivity, and delivers quicker, fairer decisions for customers.

In summary, the project shows that adopting this predictive model can help lenders cut financial losses, streamline approvals, and maintain consistency in risk management, making lending both safer and more efficient.